

Purpose: Standardizing Behavior-Driven Development for QA and Engineering.

Section 1: Prerequisites & Environment Ensure you have the Java 21 JDK installed and the `JAVA_HOME` environment variable configured. Verify Maven is installed by running `mvn -version` in your terminal.

Section 2: `pom.xml` Configuration and Dependencies Create a new Maven project. Your `pom.xml` must include the following properties and dependencies to support our Cucumber and JUnit 5 testing standards.

XML

```
<properties>
  <maven.compiler.source>21</maven.compiler.source>
  <maven.compiler.target>21</maven.compiler.target>
  <cucumber.version>7.15.0</cucumber.version>
  <junit.jupiter.version>5.10.1</junit.jupiter.version>
</properties>

<dependencies>
  <dependency>
    <groupId>io.cucumber</groupId>
    <artifactId>cucumber-java</artifactId>
    <version>${cucumber.version}</version>
    <scope>test</scope>
  </dependency>
  <dependency>
    <groupId>io.cucumber</groupId>
    <artifactId>cucumber-junit-platform-engine</artifactId>
    <version>${cucumber.version}</version>
    <scope>test</scope>
  </dependency>
  <dependency>
    <groupId>org.junit.jupiter</groupId>
    <artifactId>junit-jupiter-api</artifactId>
    <version>${junit.jupiter.version}</version>
    <scope>test</scope>
  </dependency>
</dependencies>
```

Section 3: Project Structure & First Feature

1. **Directory Architecture:** Create `src/test/resources/features` for your `.feature` files and `src/test/java/stepDefinitions` for your Java implementation code.
2. **Sample Feature (`login.feature`):**
Gherkin
 Feature: User Authentication
 Scenario: Successful login with valid credentials
 Given the user is on the login page
 When they enter valid credentials
 Then they should be redirected to the dashboard
3. **Test Runner:** Create a `RunCucumberTest.java` class in `src/test/java` annotated with `@Suite` and `@SelectClasspathResource("features")` to execute the suite.